

PHASE 2 – REQUIREMENT ANALYSIS

Functional Requirements

- Accept document type, parties, terms and effective date.
- Send user input to the backend through the /generate API.
- Generate structured legal content using the AI integration.
- Display the generated document for preview and editing.
- Allow export in TXT, DOCX and PDF formats.

Software Requirements

- Python 3.10 or later
- FastAPI and Uvicorn
- Streamlit
- Google Generative AI SDK / Gemini integration
- python-docx
- FPDF
- Pillow
- Requests
- python-dotenv

System Requirements

- Python development environment with pip.
- Internet/API access for the AI generation service.
- A web browser for accessing the Streamlit interface.

Input Requirements

- Document type
- Parties involved
- Terms and conditions
- Effective date

Output Requirements

- Generated legal-document text.
- Editable preview.
- TXT, DOCX and PDF downloadable files.
- Formatted documents with headings, terms and footer/branding support.

Non-Functional Requirements

- Simple and user-friendly interface.

- Modular separation between frontend, backend, AI and formatting modules.
- Environment variables for API-key configuration.
- Validation of incoming API data using Pydantic.

Project: LegalEase – AI-Powered Legal Document Generator | SmartBridge Phase Submission